

Weekly Progress Report - Week 01

Name: Himabindu Karri
Domain: Machine Learning
Project Name: Smart City Traffic Forecasting
Week Ending: 01
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I. Overview

During Week 1, I set up the development environment using VS Code and began working on the Smart City Traffic Forecasting project. The primary focus was understanding the dataset structure, performing data preprocessing, and conducting exploratory data analysis to identify traffic patterns and trends.

II. Achievements

1. Project Setup

- Configured Python virtual environment in VS Code.
- Installed required libraries including Pandas, NumPy, Matplotlib, Seaborn, and Scikit-Learn.
- Organized the project structure for future development.

2. Dataset Analysis

- Successfully loaded the Smart City Traffic dataset containing 48,120 records.
- Identified the main features: DateTime, Junction, Vehicles, and ID.
- Examined dataset dimensions and data types.

3. Data Preprocessing

- Checked for missing values and data quality issues.
- Converted DateTime into datetime format.
- Extracted useful features such as Year, Month, Day, Hour, and Day of Week.
- Generated a cleaned dataset for future model training.

4. Exploratory Data Analysis

- Analyzed traffic volume distribution.
- Studied traffic variations across different junctions.
- Examined hourly traffic trends and peak traffic periods.
- Generated visualizations to understand traffic behavior.

Figure 1: Traffic Distribution Analysis

- Shows the overall distribution of vehicle counts in the dataset.
- Helps identify traffic density patterns and outliers.

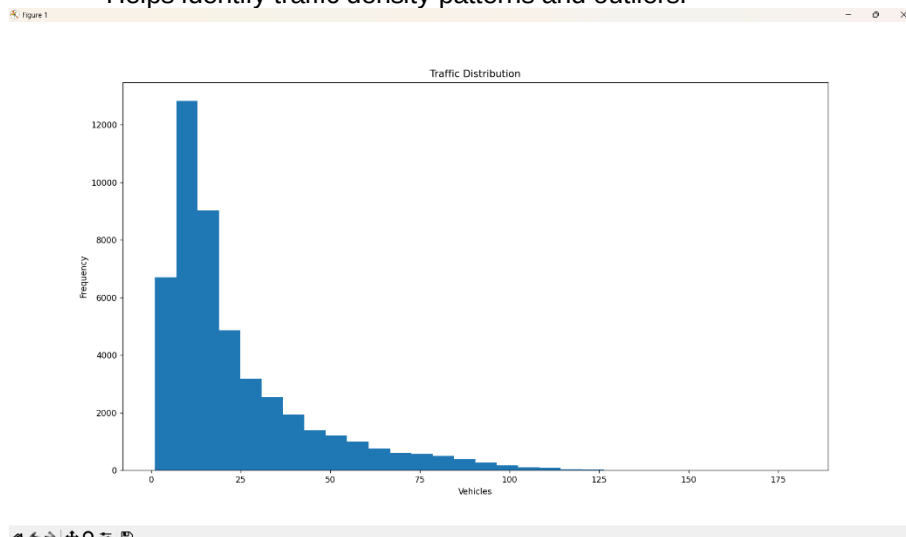


Figure 2: Average Traffic by Hour

- Illustrates hourly traffic flow patterns.
- Used to identify peak and off-peak traffic periods.

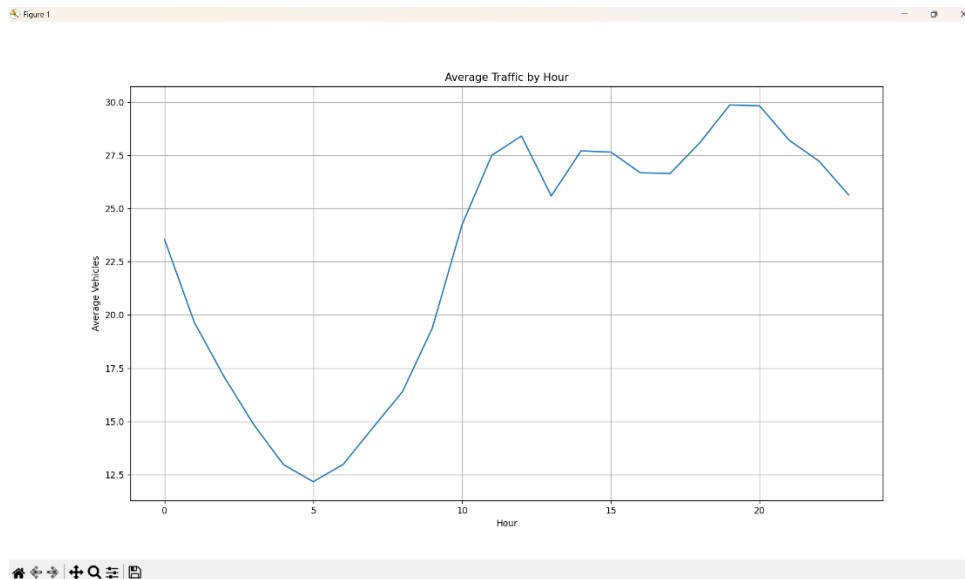
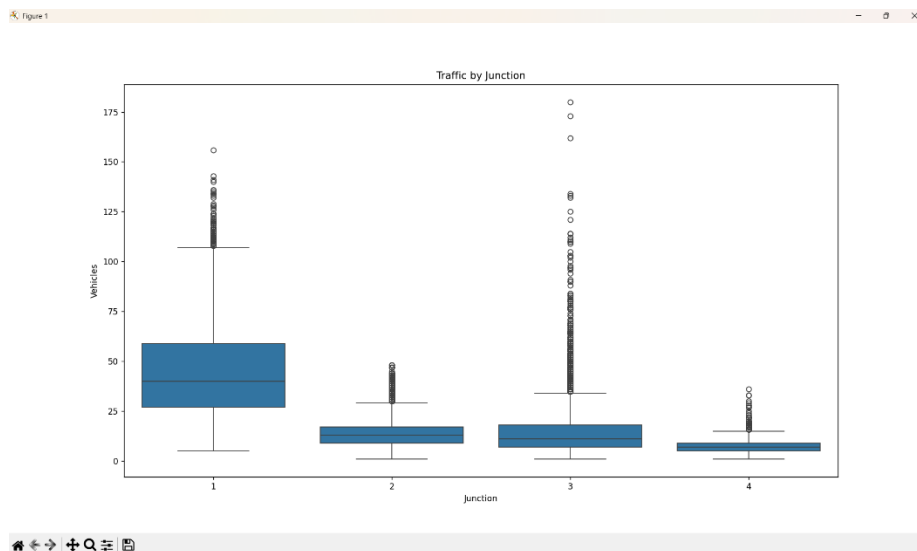


Figure 3: Traffic Variation Across Junctions

- Compares traffic volume among different junctions.
- Helps understand junction-wise traffic behaviour.



III. Challenges

- Understanding the characteristics of time-series traffic data.
- Learning effective feature extraction techniques from DateTime values.
- Organizing the project structure for long-term development.

IV. Learning Resources

- Pandas Documentation
- Matplotlib Documentation
- Seaborn Documentation
- Python Time-Series Analysis Tutorials

V. Next Week's Goals

- Perform advanced feature engineering.
- Build baseline machine learning models.
- Compare Linear Regression, Decision Tree, and Random Forest models.
- Evaluate model performance using standard metrics.

VI. Additional Comments

Week 1 provided a strong foundation for understanding the dataset and preparing it for traffic forecasting model development. The insights gained from exploratory analysis will support model building in subsequent weeks.